Does Discounting Hurt Profitability?

A Statistical Analysis of Order-Level Discount and Profit Data

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1. Executive Summary

This report asks whether the discount applied to an order affects the profit earned on it, using 9,994 order-line items from the Sample Superstore dataset. It does, but not gradually: profitability holds up well below ~30% discount (only 0–18% of orders lose money) and collapses above ~31% (89–100% lose money). That sharp break, not the discount level itself, is the actionable finding — and it's why Section 5 recommends a flat 30% approval threshold rather than a blanket discourage-all-discounting policy.

2. The Business Question

Retail businesses routinely offer discounts to close sales, but every discount comes directly out of the profit margin on that order. This raises a question every sales organisation eventually has to answer: is there a point at which discounting stops being a sales tool and starts being a loss-generating habit? This report investigates that question directly using order-level transaction data.

Formally, the question tested was: does the discount rate applied to an order have a relationship with the profit earned on that order?

Null hypothesis (H0)	There is no relationship between Discount and Profit ($\rho = 0$)
Alternative hypothesis (H1)	A relationship exists between Discount and Profit ($\rho \neq 0$)
Predictor variable	Discount — continuous, proportion (0 to 0.8)
Outcome variable	Profit — continuous, dollars (can be negative)

3. The Data

This analysis uses the Sample Superstore dataset, a widely-used retail order-level dataset covering 9,994 individual order line items placed between January 2015 and December 2018. Each row represents one product line within a customer order and includes the order date, customer segment, product category, region, discount rate, sales value, and profit.

Data quality note: the source file (a public mirror hosted on GitHub) contained 806 fully blank rows that were removed during cleaning; this was a formatting artifact of that specific file, confirmed by the cleaned row count matching the dataset's well-documented canonical size. Beyond that, only 11 records (0.1%) were missing a postal code, which is not relevant to this analysis.

Limitation: this is historical data from a single U.S. retail company. The specific 30% discount threshold identified below reflects this company's cost structure and product mix — it should be treated as a finding to investigate for this business, not a universal retail law.

4. Key Findings

Sales values are heavily right-skewed (mean $230 vs. median $54) — most orders are small, with a long tail of large ones pulling the average up.

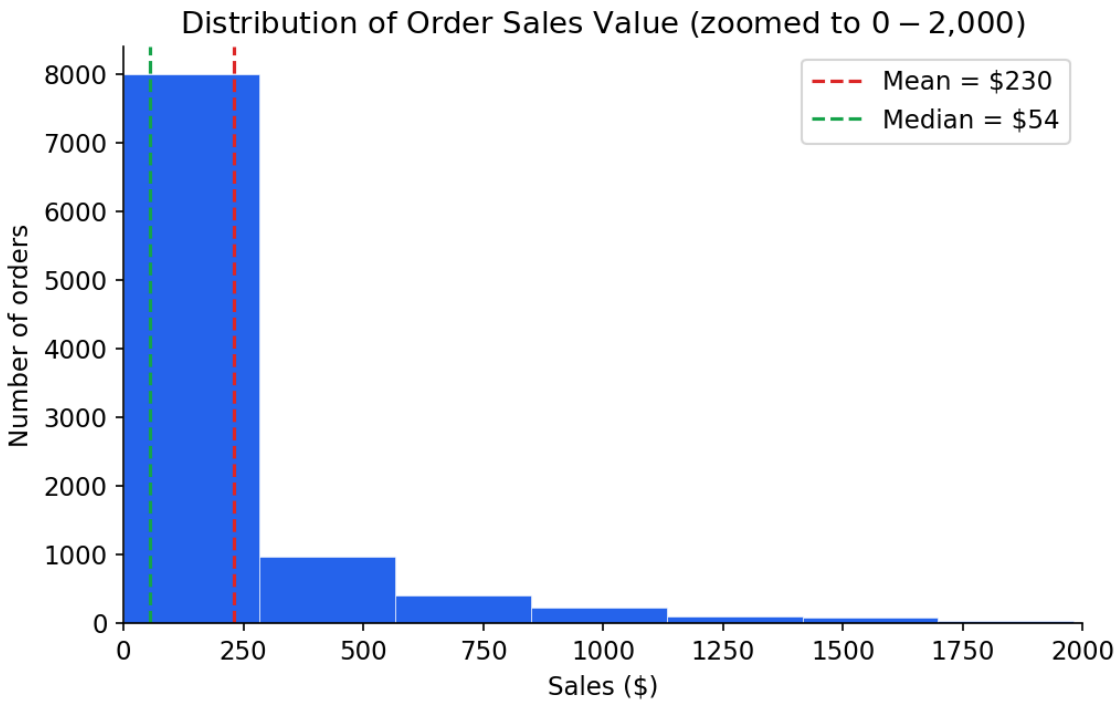


Figure 1 — Distribution of order Sales value

A first look at Discount vs. Profit shows a negative relationship, but a straight-line (Pearson) correlation understates it:

Test	Result
Pearson correlation (linear)	r = -0.2195, p < .001
Spearman correlation (monotonic)	ρ = -0.5434, p < .001
R^2 (Pearson)	0.048 — Discount alone explains 4.8% of variance in Profit

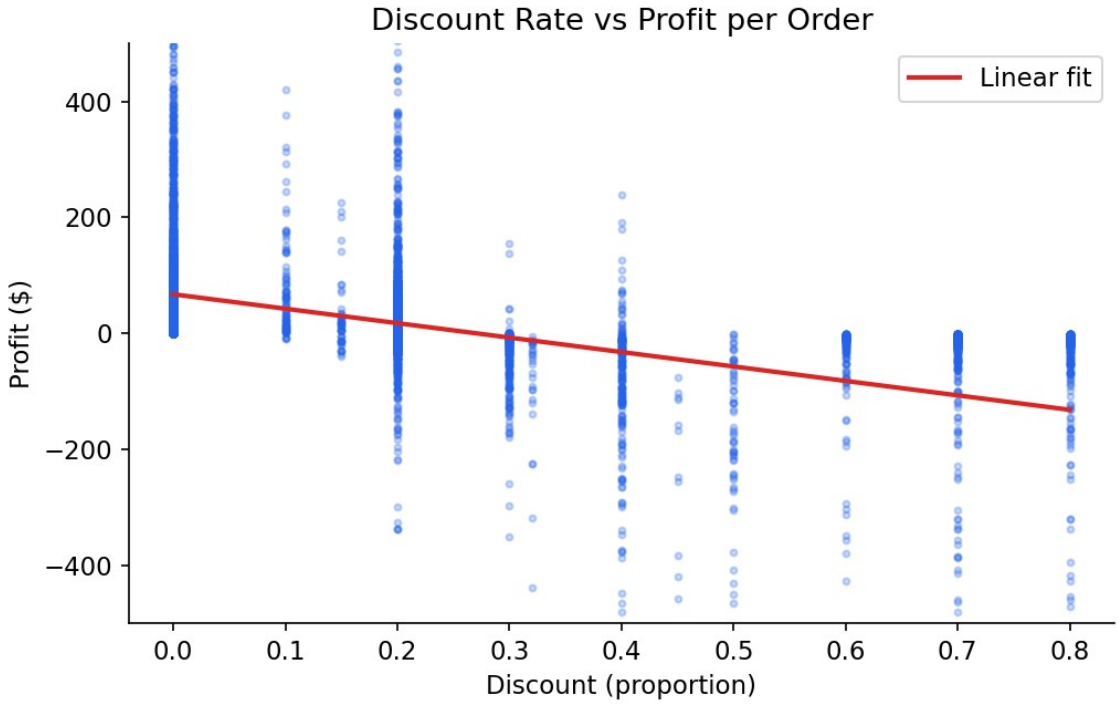


Figure 2 — Discount rate vs. Profit per order, with linear fit

The gap between the two correlation values (-0.22 vs -0.54) is itself a finding: it means Discount and Profit are related, but not in a straight line. Breaking Profit down by discount band shows why — the relationship is closer to a threshold effect than a gradual decline:

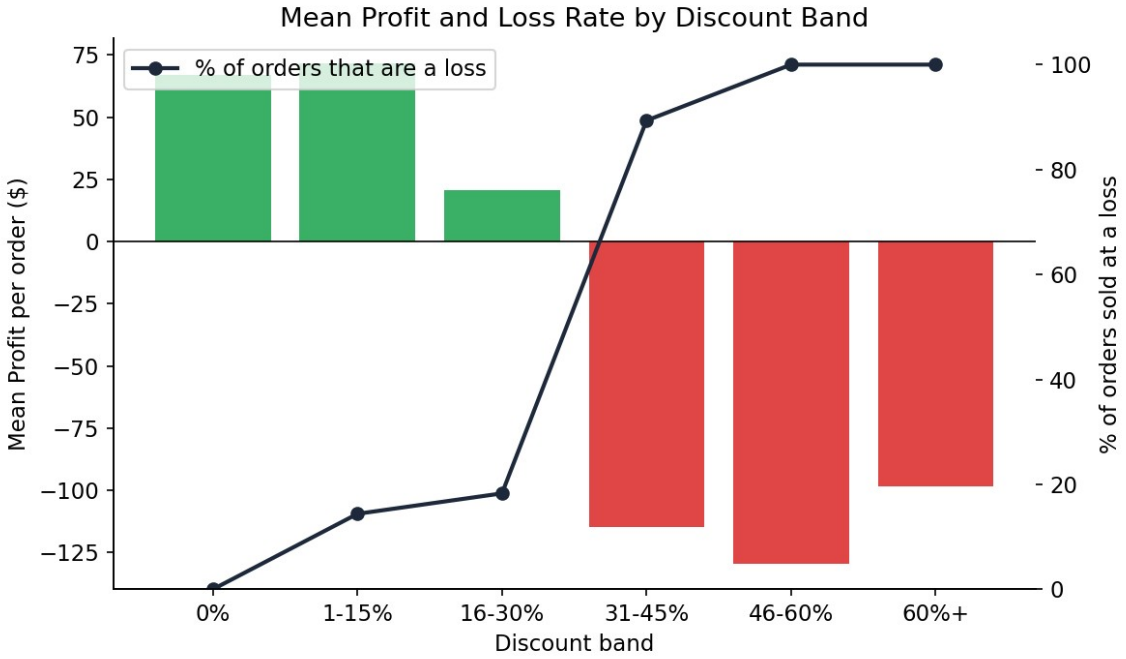


Figure 3 — Mean profit and loss rate by discount band

Discount band	Mean profit / % of orders sold at a loss
0%	$66.90 / 0.0%
1–15%	$71.56 / 14.4%
16–30%	$20.59 / 18.3%
31–45%	-$114.51 / 89.3%
46–60%	-$129.66 / 100.0%
60%+	-$98.35 / 100.0%

Below ~30% discount, the large majority of orders remain profitable. Above ~31%, almost every order is sold at a loss. This is a considerably sharper and more actionable pattern than "more discount, somewhat less profit."

Profit itself is heavily right-skewed with very heavy tails (skewness ≈ 7.6; a normal distribution is 0), driven by a mix of a few very large profitable orders and a meaningful number of high-loss orders — visible as the S-shaped departure from the reference line in the Q-Q plot below.

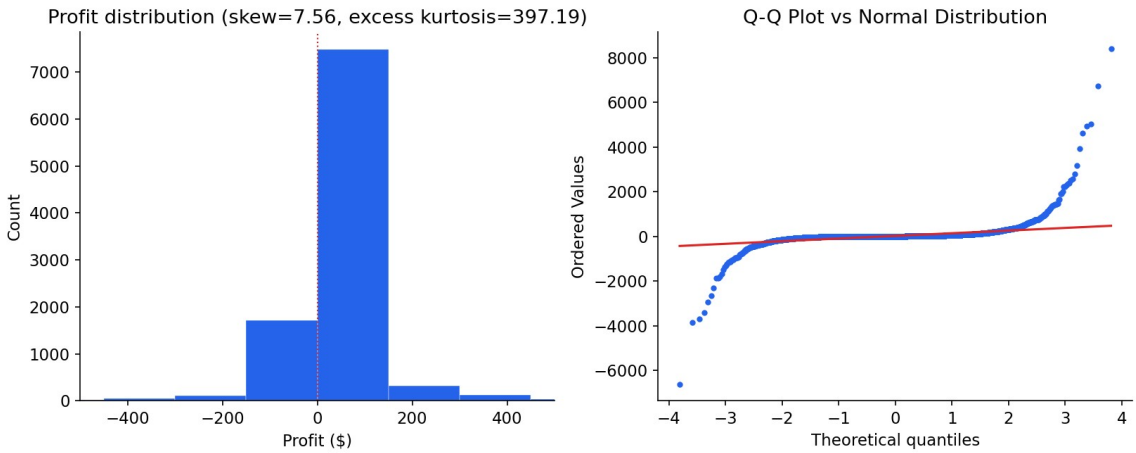


Figure 4 — Profit distribution: histogram and Q-Q plot

Assumptions & honesty check: Pearson's test assumes a linear relationship, which the discount-band breakdown shows is not quite the case here — this is why Spearman and the band analysis are reported alongside it rather than relying on Pearson r alone. Both correlations are statistically significant ($p < .001$), but with $n = 9{,}994$, even small effects reach significance; the discount-band loss-rate table is what carries the practical, decision-relevant signal, not the p-value by itself.

5. Recommendations & Next Steps

Orders discounted above 30% should require manager approval before being processed. A single, flat threshold is easy to enforce and audit — one field to check, no lookup table or category exceptions — and it lands exactly where the loss rate jumps from ~18% to ~89%. It isn't perfect: Furniture tends to stay profitable past 30%, while Office Supplies can tip into losses earlier, so a flat rule will under-restrict one and over-restrict the other. A category-specific cap (Option B) would fix that, but isn't worth the added complexity until the flat cap has run for a quarter and the team can see whether those category gaps actually cost anything in practice.

6. Appendix — Technical Details

Full code, all outputs, and additional exploratory charts (profit by category, average sales by region) are provided in the accompanying Jupyter notebook: discount_profitability_analysis.ipynb.

Item	Value
Sample size (n)	9,994 order line items
Pearson r (Discount, Profit)	-0.2195 (p = 2.70e-109)
Spearman ρ (Discount, Profit)	-0.5434 (p < .001)
Profit skewness / excess kurtosis	7.56 / 397.19
Shapiro-Wilk normality test (n=3000 subsample)	W = 0.298, p < .001 — not normal